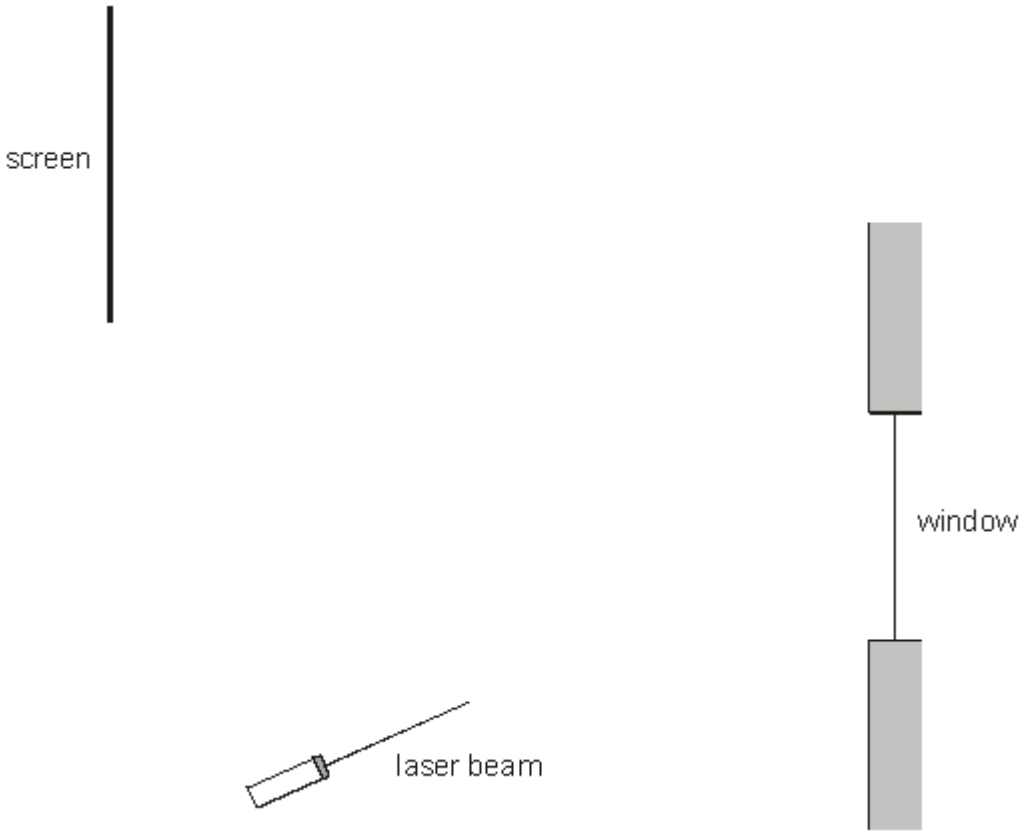
Q1.

(a) A teacher shines a laser beam onto a classroom window. It reflects off the window and onto a screen.



On the diagram above, continue the laser beam to show its path as it reflects off the window and onto the screen. Use a ruler.
Add arrows to show the direction of the laser beam.

2 marks

(b) (i) When a pupil plays her flute in the classroom the window vibrates.
Give the reason for this.

.....

.....

1 mark

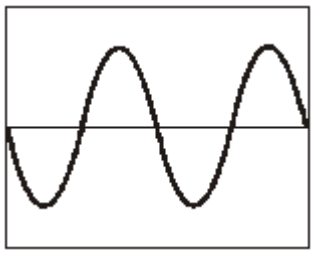
(ii) When the window vibrates, what happens to the laser beam that is reflected off the window?

.....

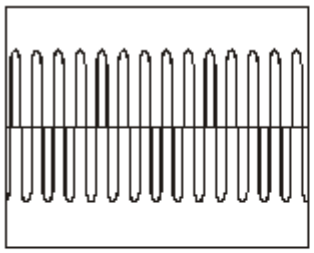
.....

1 mark

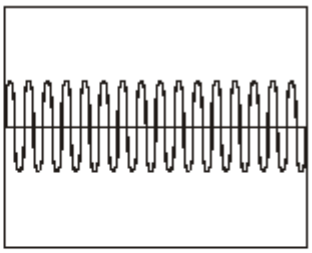
(c) The teacher places a microphone near the pupil as she plays her flute. The diagram below shows the pattern on an oscilloscope screen.



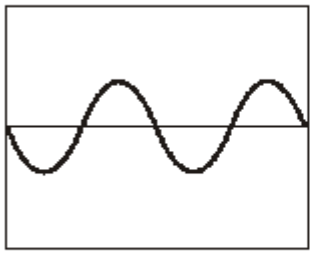
The pupil then plays her flute at a **higher pitch** and **more quietly**.
Which diagram below shows the pattern that would be seen on the oscilloscope?
Tick the correct box.



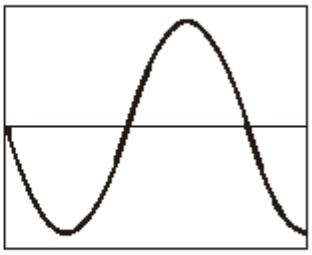
A ☐



B ☐



C ☐



D ☐

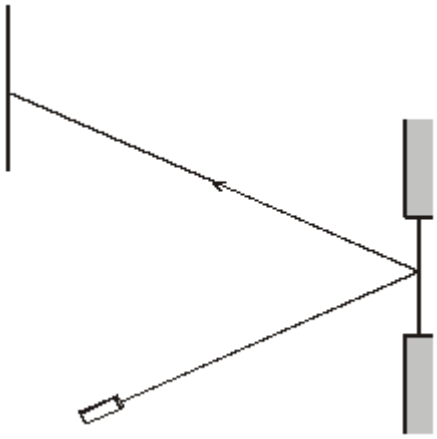
1 mark
maximum 5 marks

Q1.

(a) the angles of incidence and reflection must be equal

1 (L5)

the ray must be continuous and straight with an arrow in the correct direction



accept one arrow in the correct direction on either the incident or the reflected ray

the ray must hit the screen in the middle 2 cm

1 (L5)

(b) (i) any **one** from

- sound waves travel to the window **or** glass
  accept 'sound travels to the window'
- vibrations pass through the air
- sound waves **or** vibrations hit the window
  accept 'sound hits the window'

1 (L5)

(ii) the beam will vibrate **or** move **or** jump about

accept 'scatter' for vibrate

accept 'it would go to other places in the room'
accept 'it would go in different directions'
*accept 'it shakes' **or** 'it wobbles'*

*do **not** accept 'it is blurred'*

'it goes in a different direction' is insufficient

1 (L5)

(c) B ✓

if more than one box is ticked, award no mark

1 (L6)

[5]